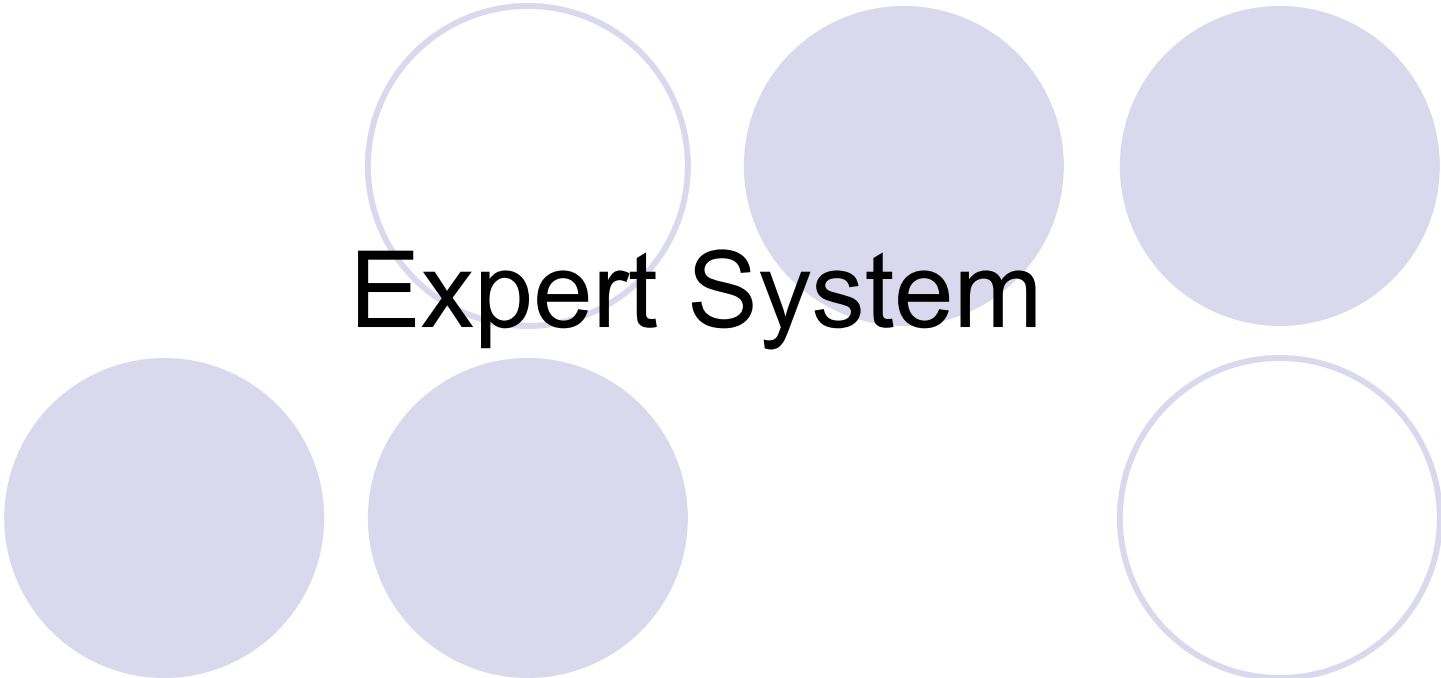




Expert System

Developing Intelligent Program Systems

Expert system

- expert system (or knowledge-based system): **a program which encapsulates knowledge from some domain, normally obtained from a human expert in that domain**
- an expert system can, to a certain extent, act as a substitute for the expert from whom the knowledge was taken.
- **Attempt to Imitate Expert Reasoning Processes and Knowledge in Solving Specific Problems**
- **Expert Systems Do Not Replace Experts, But They**
 - **Make their Knowledge and Experience More Widely Available**
 - **Permit Nonexperts to Work Better**
- **ES acts as a *Consultant* or *Advisor***

Developing Intelligent Program Systems

Objective and Activities of an Expert System

- **Objective of an expert system**
 - To transfer expertise from an expert to a computer system and
 - Then on to other humans (nonexperts)
- **Activities**
 - Knowledge acquisition
 - Knowledge representation
 - Knowledge inferencing
 - Knowledge transfer to the user
- Knowledge is stored in a *knowledge base*



Developing Intelligent Program Systems

Inferencing

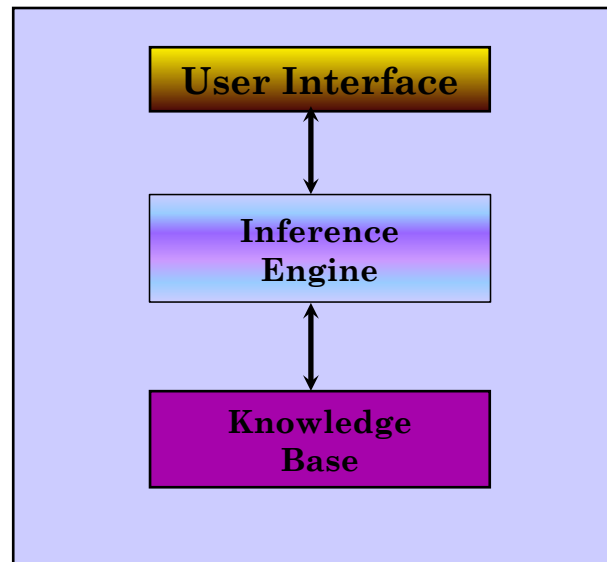
- **Reasoning** (Thinking)
- The computer is programmed so that it can make inferences
- Performed by the *Inference Engine*



Developing Intelligent Program Systems : Expert System

Major Components

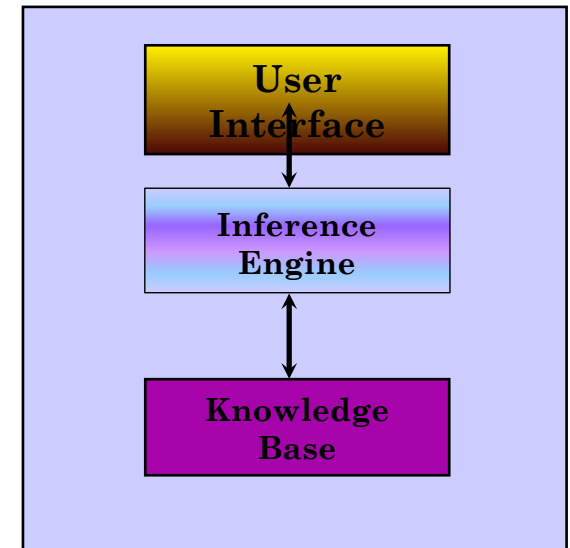
- **Knowledge base (KB)**: repository of rules, facts (productions)
- **working memory**: (if forward chaining used)
- **inference engine**: the deduction system used to infer results from user input and KB
- **user interface**: interfaces with user
- **external control + monitoring**: access external databases, control,...



Developing Intelligent Program Systems : Expert System

Knowledge base

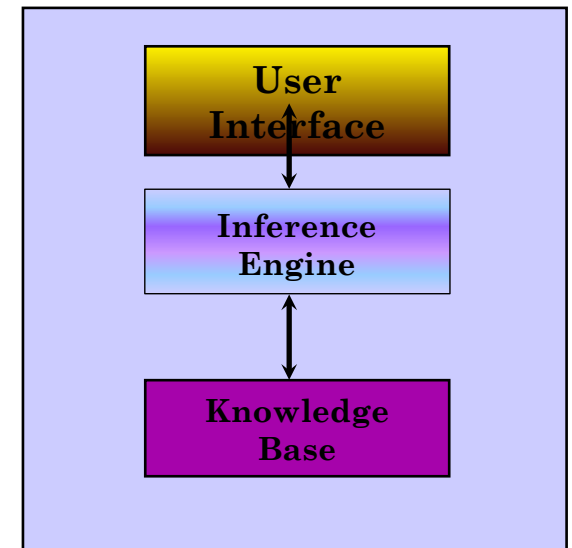
- The knowledge base contains the knowledge necessary for understanding, formulating, and solving problems
- Two Basic Knowledge Base Elements
 - Facts
 - Special heuristics, or rules that direct the use of knowledge
 - Knowledge is the primary raw material of ES
 - Incorporated knowledge representation



Developing Intelligent Program Systems : Expert System

Inference Engine

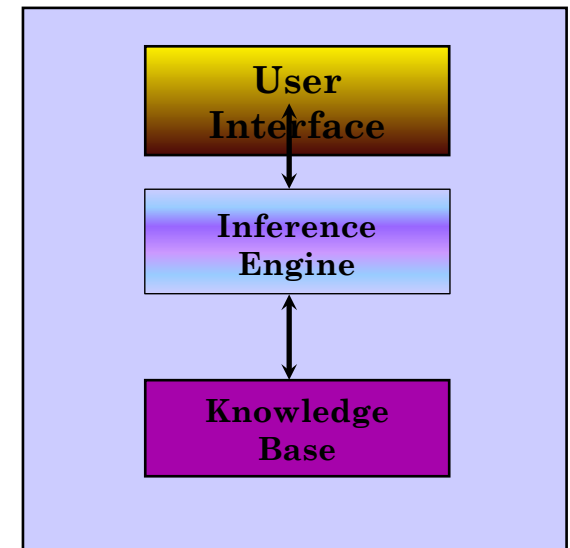
- The *brain* of the ES
- The control structure (rule interpreter)
- Provides methodology for reasoning



Developing Intelligent Program Systems : Expert System

User Interface

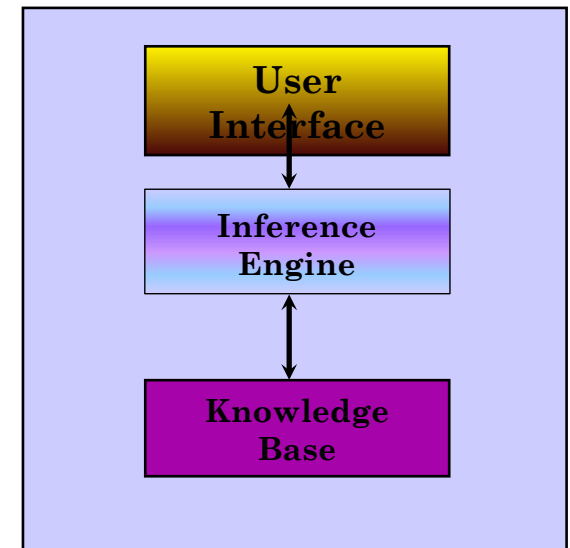
- **Language processor** for friendly, problem-oriented communication
- Natural Language Processing, or menus and graphics



Developing Intelligent Program Systems : Expert System

Working Memory (Blackboard)

- **Area of working memory to**
 - **Describe the current problem**
 - **Record Intermediate results**
- **Records Intermediate Hypotheses and Decisions**
 - 1. Plan**
 - 2. Agenda**
 - 3. Solution**



Developing Intelligent Program Systems : Expert System

Explanation Subsystem (Justifier) Area of working memory to

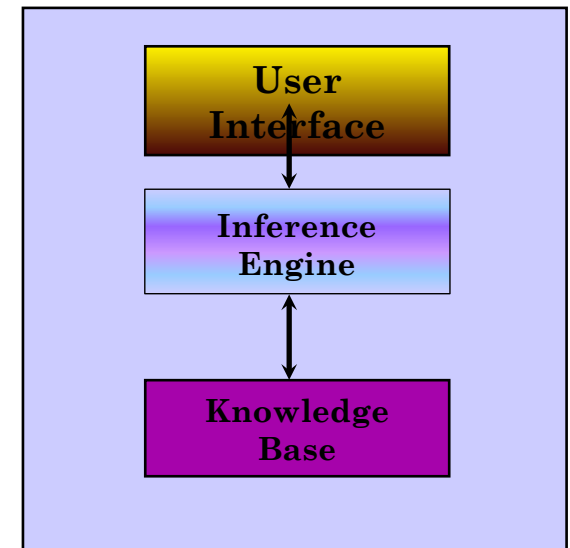
- **Traces responsibility and explains the ES behavior by interactively answering questions:**

-Why?

-How?

-What?

-(Where? When? Who?)



Developing Intelligent Program Systems : Expert System

The Human Element in Expert Systems

- **Expert**
- **Knowledge Engineer**
- **User**
- **Others**



Developing Intelligent Program Systems : Expert System

Expert

- Has the special knowledge, judgment, experience and methods to **give advice** and **solve problems**
- Provides knowledge about task performance



Developing Intelligent Program Systems : Expert System

Knowledge Engineer

- Helps the expert(s) structure the problem area by interpreting and integrating human answers to questions, drawing analogies, posing counterexamples, and bringing to light conceptual difficulties
- Usually also the **System Builder**



Developing Intelligent Program Systems : Expert System

User

- Possible Classes of Users
 - A non-expert client seeking direct advice (ES acts as a *Consultant* or *Advisor*)
 - A student who wants to learn (*Instructor*)
 - An ES builder improving or increasing the knowledge base (*Partner*)
 - An expert (*Colleague* or *Assistant*)
- The Expert and the Knowledge Engineer Should Anticipate Users' Needs and Limitations When Designing ES



Expert system

- **Examples:**

- Dendral, MYCIN, PUFF, ELIZA, BTDS, etc.

- Dendral expert system:

- The primary aim to aid organic chemists with identification of unknown organic molecules by analyzing information from mass spectrometry graphs and the knowledge of chemistry

Expert system

- MYCIN:
 - Written in LISP around 1970s and derived from Dendral expert system
 - was designed to diagnose infectious blood diseases and recommend antibiotics, with the dosage adjusted for patient's body weight
 - It would query the physician/patient running the program via a long series of simple yes/no or textual questions.
 - At the end, it provides
 - a list of possible cause bacteria ranked from high to low based on the probability of each diagnosis,
 - its confidence in each diagnosis' probability
 - the reasoning behind each diagnosis
 - It has around ~50 rules

Expert system

● PUFF:

- PUFF can diagnose the presence and severity of lung disease and produce reports for the patient's file
- Is an Expert System that interprets lung function test data and has become a working tool in the pulmonary physiology lab of a large number of hospital
- Puff went into production at Pacific Medical Center in San Francisco in 1977.
- Several hundred copies have been sold and are in use around the world.

Expert system

● ELIZA :

- ELIZA is a very well-known artificial intelligence program designed to emulate a Rogerian psychotherapist.
- The basic elements of Carl Rogers' new way of therapy was to have a more personal relationship with the patient, to help the patient reach a state of realization that they can help themselves
- ELIZA was showcased for a number of years at the **MIT AI Laboratory**.
- ELIZA has no reasoning ability, cannot learn
- ELIZA only appears to understand because "she" uses canned responses based on keywords, as well as string substitution
- Visit and interact with ELIZA at <http://www-ai.ijs.si/eliza/eliza.html>